Comparative Statics and the Concept of Derivative

ECON 320 - Introduction to Mathematical Economics

Fall 2025

Learning objectives. By the end of this Lecture you should be able to:

1. Define and interpret the derivative as a limit and as a marginal concept in economics.
2. Apply rules of differentiation (product, quotient, chain) and compute higher-order derivatives.
3. Use partial and total derivatives to analyze how endogenous variables respond to parametric changes.
4. Perform comparative statics in single-equation models: $F(x,p) = 0 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}p} = -\dfrac{F_p}{F_x}$.
5. Perform comparative statics in systems: $\mathbf{F}(\mathbf{x},\mathbf{p}) = \mathbf{0} \Rightarrow \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\mathbf{p}} = -J^{-1}B$ where $J = \frac{\partial \mathbf{F}}{\partial \mathbf{x}}$, $B = \frac{\partial \mathbf{F}}{\partial \mathbf{p}}$.
6. Use Cramer's rule for sign analysis and interpret results economically.
7. Work through canonical applications (supply–demand, tax incidence, monopoly, Cournot, Cobb–Douglas demand).

Contents

1 Why derivatives and comparative statics matter

Economic models map *assumptions* (parameters, exogenous variables) into *predictions* (endogenous variables). **Comparative statics** studies how equilibrium outcomes change when assumptions change. The **derivative** is the central tool: it converts small parameter changes into approximate outcome changes with clear sign and magnitude interpretations (*marginal effects*).

2 The univariate derivative: definition, meaning, and rules

2.1 Limit definition and economic interpretation

Definition 1 (Derivative)**.** *Let $y = f(x)$. The derivative of f at x is*

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

provided the limit exists. Economically, $f'(x)$ is the marginal *effect of x on y.*

Example 1 (Marginal cost)**.** *If total cost is $C(Q) = a + bQ + cQ^2$ with $a, b, c > 0$, then the marginal cost is $MC(Q) = C'(Q) = b + 2cQ$. MC rises with output when $c > 0$ (increasing marginal cost).*

2.2 Basic rules

For differentiable functions $u(x)$, $v(x)$ and constant k:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big[ku\big] = ku', \quad \frac{\mathrm{d}}{\mathrm{d}x}\big[u \pm v\big] = u' \pm v', \quad \frac{\mathrm{d}}{\mathrm{d}x}\big[uv\big] = u'v + uv',$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{u}{v}\right] = \frac{u'v - uv'}{v^2}, \qquad \frac{\mathrm{d}}{\mathrm{d}x} f\big(g(x)\big) = f'\big(g(x)\big) \cdot g'(x) \quad \text{(chain rule)}.$$

2.3 Higher-order derivatives, curvature, and Taylor

Definition 2. *The second derivative $f''(x)$ measures how the slope changes. If $f''(x) > 0$ locally, f is* convex*; if $f''(x) < 0$, f is* concave*.*

A second-order Taylor approximation around x_0 is

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \tfrac{1}{2} f''(x_0)(x - x_0)^2,$$

useful for local welfare and surplus approximations.

2.4 Elasticities and log-derivatives

The (point) elasticity of y with respect to x is

$$\varepsilon_{y,x} = \frac{\mathrm{d}y}{\mathrm{d}x} \cdot \frac{x}{y}.$$

If $y = f(x)$ and both are positive, $\frac{\mathrm{d} \ln y}{\mathrm{d} \ln x} = \varepsilon_{y,x}$. For power functions $y = Ax^{\alpha}$, $\varepsilon_{y,x} = \alpha$ everywhere.

3 Partial and total derivatives

3.1 Partial derivatives

For $z = f(x_1, \dots, x_n)$, the partial derivative with respect to x_i is

$$\frac{\partial z}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_n)}{h},$$

holding all other arguments fixed. Cross-partials $\frac{\partial^2 z}{\partial x_i \, \partial x_j}$ (for $i \neq j$) capture interaction effects. If f is sufficiently smooth, $\frac{\partial^2 f}{\partial x_i \, \partial x_j} = \frac{\partial^2 f}{\partial x_j \, \partial x_i}$ (Clairaut symmetry).

3.2 Total derivative and total differential

When $x = x(p)$ depends on parameter p,

$$\frac{\mathrm{d}z}{\mathrm{d}p} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} \frac{\mathrm{d}x_i}{\mathrm{d}p} + \frac{\partial f}{\partial p}.$$

Infinitesimally,

$$\mathrm{d}z = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} \, \mathrm{d}x_i + \frac{\partial f}{\partial p} \, \mathrm{d}p.$$

These formulas are the workhorses of comparative statics when equilibria are described implicitly.

4 Implicit differentiation and the implicit function theorem

4.1 Single-equation case

Consider equilibrium defined by one equation $F(x, p) = 0$, where x is endogenous and p is a parameter.

Proposition 1 (Chiang's implicit differentiation rule)**.** *If F is continuously differentiable and $F_x \neq 0$ at (x^*, p), then locally there is a differentiable function $x = x(p)$ with*

$$\boxed{\frac{\mathrm{d}x}{\mathrm{d}p} = -\frac{F_p}{F_x}}.$$

Example 2 (Linear supply–demand (parameter shift))**.** *Demand $D(p, m) = a - bp + \alpha m$, supply $S(p, w) = c + dp - \beta w$. Equilibrium: $F(p; m, w) = D(p, m) - S(p, w) = 0$. Then $F_p = -b - d < 0$, $F_m = \alpha > 0$, $F_w = -\beta < 0$. Hence*

$$\frac{\mathrm{d}p^*}{\mathrm{d}m} = -\frac{F_m}{F_p} = \frac{\alpha}{b+d} > 0, \qquad \frac{\mathrm{d}p^*}{\mathrm{d}w} = -\frac{F_w}{F_p} = -\frac{-\beta}{b+d} = -\frac{\beta}{b+d} < 0.$$

Income raises the equilibrium price (normal good), higher input costs shift supply left and lower *the price here because w enters supply with a negative sign in F; if one defines $F = S - D$ instead, signs flip consistently. Always check which side of the equation your parameter enters.*

4.2 System case and Jacobians

Let $\mathbf{x} \in \mathbb{R}^n$ solve the n-equation system $\mathbf{F}(\mathbf{x}, \mathbf{p}) = \mathbf{0}$. Define the Jacobian with respect to endogenous variables

$$J = \frac{\partial \mathbf{F}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_n}{\partial x_1} & \cdots & \frac{\partial F_n}{\partial x_n} \end{bmatrix}, \qquad B = \frac{\partial \mathbf{F}}{\partial \mathbf{p}}.$$

If J is nonsingular at the point of interest, then by the implicit function theorem

$$\boxed{\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\mathbf{p}} = -J^{-1}B}.$$

Cramer's rule for signs (2×2 illustration). With two equations and one scalar parameter p,

$$\frac{\partial \mathbf{F}}{\partial \mathbf{x}} = \begin{bmatrix} F_{1x_1} & F_{1x_2} \\ F_{2x_1} & F_{2x_2} \end{bmatrix}, \quad B = \begin{bmatrix} F_{1p} \\ F_{2p} \end{bmatrix}, \quad \Delta = \det(J) = F_{1x_1}F_{2x_2} - F_{1x_2}F_{2x_1} \neq 0,$$

$$\frac{\mathrm{d}x_1}{\mathrm{d}p} = -\frac{\det\left(\begin{matrix} \boxed{F_{1p}} & F_{1x_2} \\ \boxed{F_{2p}} & F_{2x_2} \end{matrix}\right)}{\Delta} = -\frac{F_{1p}F_{2x_2} - F_{2p}F_{1x_2}}{\Delta}, \qquad \frac{\mathrm{d}x_2}{\mathrm{d}p} = -\frac{\det\left(\begin{matrix} F_{1x_1} & \boxed{F_{1p}} \\ F_{2x_1} & \boxed{F_{2p}} \end{matrix}\right)}{\Delta} = -\frac{F_{1x_1}F_{2p} - F_{2x_1}F_{1p}}{\Delta}.$$

Signs follow from the signs of cofactors and Δ.

5 Worked applications

5.1 Tax incidence in linear supply–demand

Let inverse demand $p = \alpha - \beta Q$ ($\beta > 0$) and inverse supply $p = \gamma + \delta Q$ ($\delta > 0$). Impose a per-unit *consumer* tax t, so consumers pay $p_c = p + t$ while producers receive p. Equilibrium satisfies

$$\alpha - \beta Q = (\gamma + \delta Q) + t \quad \Rightarrow \quad Q^*(t) = \frac{\alpha - \gamma - t}{\beta + \delta}.$$

Comparative statics:

$$\frac{\mathrm{d}Q^*}{\mathrm{d}t} = -\frac{1}{\beta + \delta} < 0, \qquad \frac{\mathrm{d}p^*}{\mathrm{d}t} = \frac{\delta}{\beta + \delta} \in (0, 1), \qquad \frac{\mathrm{d}p_c^*}{\mathrm{d}t} = 1 - \frac{\delta}{\beta + \delta} = \frac{\beta}{\beta + \delta} \in (0, 1).$$

Price received by producers rises by a fraction $\delta/(\beta + \delta)$ of the tax; the rest is borne by consumers. Incidence falls on the relatively inelastic side (larger slope in levels).

5.2 Monopoly with linear demand and constant marginal cost

Suppose inverse demand $p = a - bQ$, $a, b > 0$, constant marginal cost $c \in (0, a)$. Profit $\pi(Q) = (a - bQ)Q - cQ - F$. FOC: $\pi'(Q) = a - 2bQ - c = 0 \Rightarrow Q^* = \dfrac{a - c}{2b}$, $p^* = \dfrac{a + c}{2}$. Comparative statics:

$$\frac{\mathrm{d}Q^*}{\mathrm{d}c} = -\frac{1}{2b} < 0, \quad \frac{\mathrm{d}p^*}{\mathrm{d}c} = \frac{1}{2} > 0, \quad \frac{\mathrm{d}Q^*}{\mathrm{d}a} = \frac{1}{2b} > 0, \quad \frac{\mathrm{d}p^*}{\mathrm{d}a} = \frac{1}{2} > 0.$$

Interpretation: higher cost reduces output and raises price; stronger demand shifts increase both.

5.3 Cournot duopoly (best-response system and Jacobian)

Two firms choose quantities q_1, q_2, price $p = A - B(q_1 + q_2)$, constant cost c_i. FOCs:

$$F_1(q_1, q_2; c_1) = A - B(q_1 + q_2) - Bq_1 - c_1 = 0 \;\Rightarrow\; A - c_1 - 2Bq_1 - Bq_2 = 0,$$
$$F_2(q_1, q_2; c_2) = A - c_2 - Bq_1 - 2Bq_2 = 0.$$

Jacobian $J = \begin{bmatrix} -2B & -B \\ -B & -2B \end{bmatrix}$, $\Delta = 3B^2 > 0$. Parameter matrix for c_1: $B_{(c_1)} = \begin{bmatrix} -1, & 0 \end{bmatrix}^\top$. By Cramer's rule,

$$\frac{\mathrm{d}q_1^*}{\mathrm{d}c_1} = -\frac{\det\begin{pmatrix} \boxed{-1} & -B \\ \boxed{0} & -2B \end{pmatrix}}{\Delta} = -\frac{2B}{3B^2} = -\frac{2}{3B} < 0, \qquad \frac{\mathrm{d}q_2^*}{\mathrm{d}c_1} = -\frac{\det\begin{pmatrix} -2B & \boxed{-1} \\ -B & \boxed{0} \end{pmatrix}}{\Delta} = -\frac{-B}{3B^2} = \frac{1}{3B} > 0.$$

An increase in firm 1's cost reduces its own output and (strategic substitutability) raises its rival's.

5.4 Cobb–Douglas demand and elasticities

Utility $u(x, y) = x^\alpha y^{1-\alpha}$ with $0 < \alpha < 1$; prices p_x, p_y; income I. Marshallian demand:

$$x^* = \frac{\alpha I}{p_x}, \qquad y^* = \frac{(1-\alpha)I}{p_y}.$$

Comparative statics:

$$\frac{\partial x^*}{\partial I} = \frac{\alpha}{p_x} > 0, \quad \frac{\partial x^*}{\partial p_x} = -\frac{\alpha I}{p_x^2} < 0, \quad \frac{\partial x^*}{\partial p_y} = 0.$$

Elasticities: $\varepsilon_{x,I} = 1$ (homothetic), $\varepsilon_{x,p_x} = -1$, and cross-price elasticity 0 (gross complements/substitutes determined by other preferences, here it is zero).

5.5 A general single-equation nonlinearity

Let equilibrium be $F(x, p) = \ln x + x^2 - p = 0$ with $x > 0$. Then

$$F_x = \frac{1}{x} + 2x > 0, \qquad F_p = -1.$$

Hence

$$\frac{\mathrm{d}x^*}{\mathrm{d}p} = -\frac{F_p}{F_x} = \frac{1}{\frac{1}{x^*} + 2x^*} = \frac{x^*}{1 + 2(x^*)^2} \in (0, \tfrac{1}{2}),$$

so x^* rises with p but at a diminishing rate.

6 How-to guide for comparative statics

Recipe.

1. *Write* the equilibrium conditions $F(\cdot) = 0$ (or $\mathbf{F} = \mathbf{0}$) with a clearly labeled endogenous vector $\mathbf{x}$ and parameters $\mathbf{p}$.
2. *Differentiate* implicitly: single equation $\frac{\mathrm{d}x}{\mathrm{d}p} = -F_p/F_x$; system $\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\mathbf{p}} = -J^{-1}B$.
3. *Sign* the derivatives using economically plausible slope conditions (e.g., $D_p < 0$, $S_p > 0$) and stability ($|J| \neq 0$, often $|J| > 0$).
4. *Interpret* magnitudes with elasticities or linearization; check units.
5. *Sanity checks*: limit cases (perfectly inelastic/elastic), symmetry, and dimensions.

7 Connections and extensions

Envelope theorem (preview). When an optimizer chooses x to maximize $f(x, p)$, the effect of p on the optimal value is $\frac{\partial f}{\partial p}$ evaluated at the optimum (no need to track $\frac{\mathrm{d}x^*}{\mathrm{d}p}$). This complements, but does not replace, comparative statics of x^*.

Log-linear comparative statics. If $x^*(p) > 0$, then $\frac{\mathrm{d}\ln x^*}{\mathrm{d}p} = \dfrac{1}{x^*}\frac{\mathrm{d}x^*}{\mathrm{d}p}$ and $\frac{\mathrm{d}\ln x^*}{\mathrm{d}\ln p} = \dfrac{p}{x^*}\frac{\mathrm{d}x^*}{\mathrm{d}p}$, useful for elasticity interpretations.

8 Common mistakes

- Confusing *partial* with *total* derivatives: $\frac{\partial y}{\partial p}$ holds other arguments fixed; $\frac{\mathrm{d}y}{\mathrm{d}p}$ accounts for induced changes.
- Forgetting the sign of F_x: in $-F_p/F_x$, the denominator is the slope of the equilibrium condition with respect to the endogenous variable.
- Mixing definitions of F: switching between $D - S$ and $S - D$ flips signs consistently; pick one and stick to it.
- Ignoring singular Jacobians: if $|J| = 0$, the local mapping may not exist; revisit model or check stability assumptions.

Summary

- Derivative $f'(x)$: marginal effect; rules of differentiation; curvature via $f''(x)$.
- Partial vs total derivatives; elasticities via log-derivatives.
- Implicit differentiation: single equation $\frac{\mathrm{d}x}{\mathrm{d}p} = -F_p/F_x$.
- Systems: $\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\mathbf{p}} = -J^{-1}B$; for 2×2, use Cramer's rule to sign effects.
- Applications: tax incidence, monopoly, Cournot, Cobb–Douglas.

Practice (with brief answers)

1. Let $F(x, p) = x^3 - p = 0$. Compute $\frac{\mathrm{d}x}{\mathrm{d}p}$ at $p > 0$.
   Answer: $F_x = 3x^2$, $F_p = -1$, so $\frac{\mathrm{d}x}{\mathrm{d}p} = 1/(3x^2) = 1/(3p^{2/3}) > 0$.

2. IS–LM toy: $Y = C_0 + c(Y - T) + I_0 - \alpha r + G$ and $M/P = kY - hr$. Solve for $\frac{\mathrm{d}Y}{\mathrm{d}G}$ sign using Jacobians with $x_1 = Y, x_2 = r$.
   Answer (sign): Under $0 < c < 1$, $h, \alpha, k > 0$, $|J| > 0$ and $\frac{\mathrm{d}Y}{\mathrm{d}G} > 0$.

3. Cournot with symmetric costs c. Show $\frac{\mathrm{d}q_1^*}{\mathrm{d}A} > 0$, $\frac{\mathrm{d}q_1^*}{\mathrm{d}B} < 0$.
   Answer: From closed forms $q_i^* = \dfrac{A - c}{3B}$, derivatives follow immediately.

References

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